CHAPTER TWO

Is Fine-Tuning Evidence that God Exists?

Teleological arguments point to some feature of the natural world and claim that, given this feature, it is probable that the natural world is the product of an intelligent designer rather than mere chance. The topic of this chapter is fine-tuning – roughly, the fact that conditions in our universe had to be almost exactly as they are in order for it to be possible for creatures like ourselves to come to exist in it – and whether the fact that the natural world has this feature provides evidence that a designer (God) exists. Robin Collins argues that it does, while Elliott Sober argues that when we make use of the relevant probabilities, we can see that fine-tuning provides no support at all that the source is God rather than mere chance.

Fine-Tuning is Evidence that God Exists

Robin Collins

Since the 1970s, physicists have discovered that the basic structure of the universe must be adjusted to an extraordinarily high degree of precision in order for complex life to exist. This discovery is called the *fine-tuning of the universe for life* or *anthropic fine-tuning*. In this essay, I will present the basic argument for why this fine-tuning provides strong evidence for theism. To do this, we first must look at the evidence for fine-tuning.

I. Evidence for Fine-Tuning

When considering the fine-tuning as evidence for theism, the relevant kind of life is what I call *embodied conscious agents* (ECAs), defined as conscious beings that substantially interact with each other for good or ill and can make choices based on what they

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consider ethical criteria. Thus, whenever I use the word "life," I am referring to such beings. The fine-tuning argument assumes that the existence of such beings requires stable, reproducible complexity.

The evidence for fine-tuning falls into three basic categories:[1] (1) that of the laws and principles of physics that govern the universe, such as the law of gravity and electromagnetism; (2) that of the constants of physics; and (3) that of the initial conditions of the universe. In this essay, I will only consider (2) and (3).

The constants of physics are the fundamental numbers that when plugged into the laws of physics determine the basic structure of the universe. The laws specify how one quantity varies with another, with the exact magnitude of variation being determined by one or more of these constants. As our first example, consider Isaac Newton's gravitational constant G that occurs in his law of gravity, $F = \frac{Gm1m2}{r2}$. This equation determines the force, F, of gravity between any mass pair, m_1 and m_2, separated a distance r apart. As can be seen by the above equation, knowing the values of the masses and the distance they are apart is not enough to determine the force; one also needs to know the value of G. Each of the other forces also has its own constant that plays a similar role to G in determining the force's strength.[2]

As our first illustration of this fine-tuning, consider the effect of increasing G by a billion-fold. This would cause the force of gravity on a planet with the mass and size of the earth to be a billion times greater, so great that organisms anywhere near the size of human beings, whether land-based or aquatic, would be crushed. Even a much smaller planet of only 400 feet in diameter – which is not large enough to sustain organisms of our size – would have a gravitational pull of one thousand times that of earth, still too strong for organisms of our brain size, and hence level of intelligence, to exist. As astrophysicist Martin Rees notes, "In an imaginary strong gravity world, even insects would need thick legs to support them, and no animals could get much larger."[3]

To determine the degree of fine-tuning of G, we must compare this billion-fold increase to the total range of force strengths in nature. Using a standard measure of force strengths – which turns out to be roughly the relative strength of the various forces between two protons in a nucleus – gravity is the weakest of the forces, and the strong nuclear force is the strongest, being a factor of around 10^{40} – or ten thousand billion, billion, billion, billion – times stronger than gravity. Consequently, compared to the total range of the strengths of the forces in nature, a billion-fold increase in the strength of gravity is tiny, being one part in ten thousand, billion, billion, billion of the entire range.

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[1] See Collins, R. (2009). The Teleological Argument: An Exploration of the Fine-Tuning of the Universe. In W.L. Craig and J.P. Moreland (ed.), *The Blackwell Companion to Natural Theology*. Oxford: Wiley Blackwell, pp. 202–281. Especially pp. 211–213.

[2] Technical treatments of the fine-tuning typically use versions of these constants that do not have units (e.g. mass, length, time, etc.). These are called *dimensionless* constants. A standard dimensionless version of G is the gravitational fine-structure constant, $\alpha_G \equiv G\frac{m_p^2}{\hbar c}$, where m_p is the mass of the proton, $\hbar$ is the reduced Planck's constant, and c is the speed of light.

[3] Rees, M. (2000). *Just Six Numbers: The Deep Forces that Shape the Universe*. New York: Basic Books, p. 30. When one takes into account the requirement that a planet retain an atmosphere, the allowed increase in G is much, much smaller than this.

The most discussed case in the physics literature of fine-tuning is that of the cosmological constant. The cosmological constant is a number in Einstein's law of gravity that, along with other factors, determines the expansion rate of space. If it were too large, space would expand so rapidly after the universe began to exist that galaxies and stars could not form; and if too small, the universe would collapse before life could evolve. The fine-tuning of the cosmological constant is typically estimated to be one part in 10^{120} (10^{120} is 1 followed by 120 zeroes, an enormously large number). This is far more precision than would be required to hit a bull's-eye the size of an atom by throwing a dart from the moon at the surface of the earth!

Further, besides the forces in nature, there are many other fundamental constants in nature, such as the masses of the quarks that compose the proton and neutron. In order for life to exist, many of these must fall into an extremely narrow range of values compared to their theoretically possible ranges.

The most impressive case of fine-tuning is that of the initial configuration of mass-energy of the universe. To get a universe like ours that allows for life, the configuration must fall within an exceedingly small region of what physicists call the *phase space* of the universe, which is just the space of possible configurations of mass-energy. According to Roger Penrose, one of Britain's leading theoretical physicists, "In order to produce a universe resembling the one in which we live, the Creator would have to aim for an absurdly tiny volume of the phase space of possible universes."[4] How tiny is this volume? According to Penrose, it is one part in $10^{10^{123}}$ – that is, 10 raised to the 10^{123} power – of the entire volume. To get an idea of this level of precision, take a sheet of paper and fill the paper with zeroes. Now, duplicate that paper enough times to fill the entire visible universe (which is approximately 14 billion light years in radius). If you were to place a "1" in front of all the zeroes on all those sheets of paper combined, the number you would get would be vastly smaller than 10 raised to the 10^{123} power. This is a truly enormous degree of fine-tuning.

A. Some Clarifications

Before looking at the possible explanations of the fine-tuning, several clarifications should be made concerning evidence cited above. First, a few physicists have called into question some cases of fine-tuning in the literature. However, even if some of the cases of fine-tuning were found to be flawed, the wide variety of independent cases should make us very confident that the universe is indeed fine-tuned for life. The situation is analogous to a court case in which the fingerprint evidence, the DNA evidence, the testimony of many witnesses, emails found on the defendant's computer, and so forth all point to the defendant's guilt. Even if one or two pieces of the evidence were found to be faulty – such as the testimony of one of the witnesses – we could be quite confident that the defendant was guilty.[5]

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[4] Penrose, R. (1989). *The Emperor's New Mind: Concerning Computers, Minds, and the Laws of Physics*. Oxford: Oxford University Press, p. 343.

[5] The most commonly cited critic of the fine-tuning is the late physicist Victor Stenger. Elsewhere, I (and others) have pointed out the many flaws in his criticism of the fine-tuning evidence. Specifically, see Collins, R. (2013). The Fine-Tuning Evidence is Convincing. In J.P. Moreland, C. Meister, and K.A. Sweis (ed.), *Debating Christian Theism*. New York: Oxford University Press, pp. 35–46. For a recent fairly thorough and accessible treatment of the fine-tuning evidence, see Barnes, L. and Lewis, G. (2016). *A Fortunate Universe: Life in a Finely Tuned Cosmos*. Cambridge: Cambridge University Press. Also see Rees, op. cit.

Second, some have wondered whether the fine-tuning evidence would be undermined if some new theory were developed whose fundamental principles and laws somehow *required* that the constants of physics have life-permitting values. Development of such a theory would indeed eliminate the fine-tuning of the constants as an independent category of fine-tuning, but it would not eliminate the fine-tuning of the laws and principles; the fine-tuning of the constants would in effect be transferred to the fine-tuning laws and principles. As astrophysicists Bernard Carr and Martin Rees note, "even if all apparently anthropic coincidences could be explained [in terms of such a unified theory], it would still be remarkable that the relationships dictated by physical theory happened also to be those propitious for life."[6]

Finally, some have objected that as far as we know there might be many other radically different law structures that allow for life. The idea is that if many radically different potential law structures would allow for life, then the fact that the actual law structures also allow for life does not require an explanation. To see the problem with this objection, consider an analogy. I open up the hood of a car and notice that the parts all have to be adjusted just right for it to function, with the vast majority of possible arrangements leading to a car that does not work. The fact that there are vehicles with very different kinds of parts does not take away from the fact that the parts in the particular car in question must be arranged just right to function, and hence from the inference that it was designed. Similarly, the potential existence of radically different law structures that allow for life does not take away from the fact that our universe must have just the right laws, constants, and initial conditions for life.

II. Evidence for Theism

A. Main Inference Pattern

The evidence physicists have gathered is that the life-permitting initial conditions and values of the constants of physics must be in a very small region relative to the range of values for which we can make reasonable determinations as to whether the universe is life-permitting. I will refer to this as the *fine-tuning evidence*. The argument in this section will be that the fine-tuning evidence provides strong confirmation of *theism* over what I call the *naturalistic single-universe hypothesis*, by which I mean that all that exists is a single universe that exists as a brute, unexplained fact. The major alternative response to the fine-tuning evidence is the *multiverse hypothesis*, which purports to explain the existence of a fine-tuned universe. We will address this alternative in the next major section below.

The following analogy illustrates the way in which the fine-tuning evidence supports theism over the naturalistic single-universe hypothesis. Suppose there is a lottery in which a set number of tickets is produced, with a person only being allowed to buy one ticket. Further, suppose the lottery commissioner's sister won. You are part of the grand

[6] Carr, B. and Rees, M. (1979). The Anthropic Cosmological Principle and the Structure of the Physical World. *Nature* 278: 605–612,

jury to determine whether to indict the lottery commissioner for rigging the lottery in favor of his sister. Initially, you are mistakenly told that it is a small, local lottery (say 100 lottery tickets were sold). You judge that the sister's winning is not that unlikely, and so it is not worth further investigating. However, at the end of the first day of trial, you are told that one-hundred million tickets were sold, and the chance of any particular ticket being the winning number is one in a hundred million. Clearly, this new information should greatly increase your judgment of the likelihood that the commissioner rigged the lottery.

Notice that your new information is not that the commissioner's sister won, but rather that having a winning ticket is far, far rarer than you previously thought. Similarly, in the case of the fine-tuning, we knew that life exists long before the discovery of fine-tuning; hence the new evidence can be thought of as evidence that the life-permitting values of the constants are far rarer than people previously thought. Further, in analogy to the lottery case, this new information should greatly increase one's judgment of the likelihood of theistic creation – that the constants were "rigged" to have the values they do.

What drives the argument in the above lottery case is that (1) under the chance hypothesis, the probability of the sister winning becomes smaller in proportion to the ratio of the number of winning tickets to the total number of tickets produced; and (2) we judge that the probability that the lottery commissioner would rig it in favor of his sister either does not depend on the number of tickets produced or does not fall as rapidly with an increase in the number of tickets produced as under the chance hypothesis. What makes (2) true is that we can discern some special motivation for the commissioner rigging it in favor of the person who won; this independent motivation keeps our judgment of the probability under the rigging hypothesis from decreasing by the same factor as the chance hypothesis as rarity of holding the winning ticket increases. To see this, consider a case in which some person with no known connection to the lottery commissioner won. Then, given that the commissioner rigged the lottery, the chance of the lottery commissioner rigging it in favor of that person (instead of any of the other lottery-ticket buyers) would become smaller in proportion to the ratio of winning tickets to the number of tickets produced.

As in the lottery case, what drives the fine-tuning argument is the claim that (i) the probability that a life-permitting universe exists under the naturalistic-single universe hypothesis is smaller in proportion to the rarity of life-permitting initial conditions and constants of physics; and (ii) the probability that God would want to create a life-permitting universe does not get smaller in proportion to this rarity. And, like the lottery commissioner case, (ii) is justified by our being able to discern some special motivation for God's creating a universe that is life-permitting. Since God is perfectly good, we would expect God to create a reality that positively, if not optimally, realizes value. Hence, as long as we can discern some positive value in the existence of embodied conscious agents, we can discern such a motivation.

However, there is a crucial difference between the lottery case and that of the fine-tuning argument. In the lottery case, we can justify the probability claim for the chance hypothesis – namely, claim (1) above – either experimentally or theoretically: for example, by having the lottery ticket machine produce many, many tickets and determining whether it truly produces numbers at random, or by deriving the probability from a

theory about the way the machine is constructed. Neither of these is possible for the analogous claim – given by (i) above – in the case of the fine-tuning argument. As I will now argue, however, the claims of probability in the fine-tuning argument can be justified in a similar way as commonly occurs in science.

B. Justifying Probabilities in the Fine-Tuning Argument

To justify the claims of probability in the case of the fine-tuning argument, we must get clear on the kind of probability in the argument, which I think is best understood as what philosophers call "epistemic probability." Although there are several accounts of epistemic probability, the one I prefer is the first major account of this kind of probability offered by John Maynard Keynes in 1921 in his *A Treatise on Probability*, an account that I have developed further elsewhere.[7] According to this account, the *conditional epistemic probability* of a proposition A on a proposition B – written P(A|B) – is to be understood as the rational degree of support that B *of itself* gives to A. Typically, we determine P(A|B) by "bracketing out" all other sources of support or warrant for A or not A (and any undercutting defeaters for A). Thus, for instance, suppose one sees it raining outside but wants to access the conditional epistemic probability of "it will rain today" on "weather.com has predicted a 25% chance of rain." In assessing this conditional probability, one blocks out all other sources of support for its raining or not raining today (such as seeing dark clouds on the horizon), except for the claim regarding the forecast of weather.com. Hence, P(it will rain today | weather.com has predicted a 25% chance of rain) = 25%. In the special circumstance when one believes A and nothing else relevant to the truth of B, then P(A|B) is the degree to which one should expect B to be true – for instance, in the last example one should have a 25% expectation that it will rain today given one only knows of weather.com's prediction.

Epistemic probability is pervasive in science and everyday life. When, for instance, scientists say that it is highly probable that the theory of evolution is true, they are saying that our current evidence strongly *supports* the truth of the theory of evolution. They are not, for example, making a statement of statistical probability, such as that the relative frequency of universes in which evolution is true is far greater than those in which it is false.

In the case of fine-tuning, the relevant probabilities in the above rendering of the argument are P(LPU | NSU & FTE) and P(LPU | T_e & FTE), where LPU represents the claim that a life-permitting universe exists, T_e represents the claim that God exists and desired to create a life-permitting universe, NSU is the naturalistic-single universe hypothesis, and FTE is the evidence of fine-tuning. The two probabilities are to be understood, respectively, as the degree of support that NSU & FTE of itself and T_e & FTE of itself give to LPU.[8] To assess this support, we use what I call the *restricted principle of indifference*,

[7] See Collins (2009). op. cit., pp. 228–233.

[8] To get an intuitive handle on these probabilities, one could think of them as the respective degrees of belief in LPU that an unembodied being with cognitive faculties relevantly similar to ours should have if they believed NSU & FTE, or T_e & FTE, and nothing else relevant to whether LPU is true.

bracketing out any information as to whether LPU is the case, including knowledge of our own existence.[9] Applied to the case at hand, this principle says that one should place an equal probability distribution over the fine-tuned variable, as long as the variable can be considered a "natural variable" – in our case, a variable that corresponds to some physical quantity (such as mass) or to a parameter used in the simplest formulation of the fundamental laws of physics.[10] The use of this principle implies that probability of a parameter falling into its life-permitting range to the appropriately chosen comparison range, which elsewhere I have called the *epistemically illuminated* (EI) region. This is just the region of values for a parameter such that, holding the laws of nature and values of the other constants the same, we can be reasonably confident whether or not ECAs could exist for the values of the parameters in question.

As with the use of epistemic probability, the use of the restricted principle of indifference is pervasive in everyday and scientific inference. As Roy Weatherford notes in his book, *Philosophical Foundations of Probability Theory*, "an astonishing number of extremely complex problems in probability theory have been solved, and usefully so, by calculations based entirely on the assumption of equiprobable alternatives [that is, the principle of indifference]."[11] Further, it is often implicitly used in scientific practice. As one of many examples that could be given, consider the case of confirmation of the hypothesis of atoms. According to Wesley Salmon, what finally convinced virtually all physical scientists by 1912 of this hypothesis was the agreement of at least thirteen independent determinations of Avogadro's number based on the assumption that atomic hypothesis was correct.[12] The scientists reasoned that if this hypothesis were false, then such an agreement between these different determinations would be "utterly astonishing coincidence"[13] – that is, it would be enormously epistemically improbable. Clearly, their judgment of probability was based neither on some relative frequency (since we only have our universe to observe) nor on some theoretical derivation. Rather, it was based on an assessment that (1) the proportion of ways in which the numbers could have disagreed is vastly larger than the proportion of ways in which they could have agreed; and (2) if atomic theory is false, we have no reason to expect a combination that agrees rather than one that disagrees. Similarly, the judgment of improbability in the case of fine-tuning is based on the judgment that (i) the proportion of initial conditions and values of the constants that are not life-permitting is vastly larger than those that are life-permitting (relative to the range of values for which we can make such estimates); and (ii) the naturalistic single-universe hypothesis provides no reason to think that the universe that exists has life-permitting, instead of non-life-permitting, initial conditions and values of the constants.

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[9] See Collins (2009). op. cit., pp. 234–239 for a more detailed treatment.

[10] When there are multiple potential natural variables depending on what one considers the simplest formulation of the fundamental laws, then one can think of the probability as falling into the range given by the different choices of natural variables, or one can average the different probabilities according to the plausibility of the formulation in question being the simplest.

[11] Weatherford, R. (1982). *Foundations of Probability Theory*. Boston: Routledge and Kegan Paul, p. 35.

[12] Salmon, W. (1984). *Scientific Explanation and the Causal Structure of the World*. Princeton: Princeton University, pp. 219–220.

[13] Ibid., p. 220.

III. Objections

A. Multiverse Objection

The most common non-theist explanation of the existence of a life-permitting universe is that there are a very large number of universes, each with different values for the fundamental parameters of physics. If such multiple universes exist, it would be no surprise that the parameters in one of them would have just the right values for the existence of life, just as in the case where if enough lottery tickets were generated, it would be no surprise that one of them would turn out to be the winning number. Further, it is no surprise that we observe that *our* universe has these values, since they are necessary for our existence.

How did these universes come into existence? The typical answer is to postulate some kind of physical process, what I will call a "universe generator." Against the naturalistic version of the universe-generator hypothesis, one could argue that the universe generator itself must be fine-tuned to produce even one life-sustaining universe. After all, even a mundane item such as a bread-making machine, which only produces loaves of bread instead of universes, must have just the right mechanisms, programming, and proportion of ingredients (flour, yeast, gluten, and so on) to produce decent loaves of bread. Indeed, as I have shown in detail elsewhere, if one carefully examines the most popular and most well developed universe-generator hypothesis, that arising out of inflationary cosmology, one finds that it contains just the right fields and laws to generate life-permitting universes. Eliminate one of the fields or laws, and no life-sustaining universes would be produced. Further, it still requires extensive fine-tuning of the initial conditions.[14]

The multiverse (of any variety) runs into additional problems, which I can only briefly mention here. First, it runs into the so-called Boltzmann Brain problem, a problem named after the physicist Ludwig Boltzmann who first proposed a multiverse to explain the low entropy of the universe. The problem is that under the standard probability measure of statistical mechanics, it is vastly more likely – on the order of $10^{10^{123}}$ times as likely – for an isolated observer to exist as a result of a random fluctuation than for the universe to have a low-entropy throughout by chance. This is not controversial. For thirty years or more cosmologists have been attempting to find an explanation of this low entropy that does not require such enormous fine-tuning, so far without success.[15] However, even if they succeed, that itself will involve hypothesizing some special set of laws to do the job, and hence additional fine-tuning of the laws of nature not explained by the multiverse.

Second, the multiverse hypothesis cannot explain the level of seeming mathematical elegance and beauty of the laws of nature. This feature of the universe has been noted by prominent physicists, both theists and atheists. Nobel Prize winning physicist Steven Weinberg, for instance, devotes a whole chapter of his book *Dreams of a Final Theory* to explaining how the criteria of beauty and elegance are commonly used with

[14] For a discussion of this issue, see McCoy, C.D. (2015). Does Inflation Solve the Hot Big Bang Model's Fine Tuning Problems? *Studies in History and Philosophy of Modern Physics* 51:23–36. DOI: 10.1016/j.shpsb.2015.06.002

[15] For an accessible discussion of this problem, see Barnes and Lewis, op. cit., pp. 314–322.

great success to guide physicists in formulating the fundamental laws of physics.[16] Weinberg, who is a convinced atheist, even admits "sometimes nature seems more beautiful than strictly necessary."[17] Because the criteria of elegance has been so successful in helping develop highly predictively accurate theories, it cannot plausibly be dismissed as merely subjective. Finally, the laws of nature themselves seem to be carefully arranged so that they are intelligible, as well as discoverable by beings with our level of intelligence – like a clever puzzle for such beings to solve. This has been stressed by many prominent physicists, and some philosophers.[18] Indeed, presently I am finishing a seven-year project showing how the fundamental parameters of physics appear to be fine-tuned to optimize our ability to discover the universe.

These additional features all make sense under theism, particularly if our ability to discover the universe is of value. For example, the existence of stars and galaxies requires low entropy, and hence a universe with low entropy throughout maximizes our ability to discover the universe. Further, as noted above, the perceived beauty and elegance of the underlying laws have greatly aided our ability to discover them. Now, we generally perceive discovering the universe as a good thing, which is why governments spend billions of dollars funding science projects whose sole purpose is to discover the nature of our universe, instead of on more immediately practical projects. Consequently, it is not surprising that a perfectly good God would create a universe that has the above features.

B. Weak Anthropic Principle Objection

According to the *weak anthropic principle*, if the structure of the universe were not fine-tuned for life, we should not be here to comment on the fact, and hence we should not find it improbable at all that we exist in a fine-tuned universe since it follows from our existence. As it stands, however, this objection confuses the probability of embodied conscious agents observing that their universe is structured to allow for their existence (given that they analyze their universe's structure) with the probability that such beings exist. The former probability is 100% since their existence entails the universe is structured to allow for their existence, whereas the latter could be very small. This is brought out by the firing squad analogy offered by John Leslie and others.[19] As Leslie points out, if 50 sharpshooters all miss you, the response "if they had not missed me I would not be here to consider the fact" is inadequate. Instead, you would naturally conclude that

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[16] Weinberg, S. (1994). *Dreams of a Final Theory*. New York: Vintage Books.

[17] Ibid., p. 250.

[18] Albert Einstein, for example, famously remarked that "the eternal mystery of the world is that it is comprehensible. … The fact that it is comprehensible is a miracle." Calaprice, A. (ed.) (1996). *The Quotable Einstein*. Princeton, NJ: Princeton University Press, p. 197. Similarly, in his famous essay, "The Unreasonable Effectiveness of Mathematics in the Physical Sciences," Eugene Wigner, one of the principal founders of quantum mechanics, famously claimed that "The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve." Wigner, E. (1960). The Unreasonable Effectiveness of Mathematics in the Physical Sciences. *Communications on Pure and Applied Mathematics* 13: 14.

[19] Leslie, J. (1989). How to Draw Conclusions from a Fine-Tuned Cosmos. In R. J. Russell, W. R. Stoeger, G. V. Coyne (ed.), *Physics, Philosophy and Theology: A Common Quest for Understanding*. Vatican City State: Vatican Observatory Press, pp. 297–312. Especially p. 304.

there was some reason why they all missed, such as that they never really intended to kill you. Why? Because your existence seems very improbable on the hypothesis that they intended to kill you, but not on the hypothesis that they did not. And this is true even though the probability is 100% that, given you continue to exist after they have shot, you will observe that they missed (given we discount the possibility of you continuing to exist after death).

In philosophy, the most prominent advocate of this objection has been Elliott Sober.[20] In response to the first point above, Sober claims that one should include one's own existence in one's background information when formulating probabilities. Given this background information, the probability that there is a life-permitting universe given the naturalistic single-universe hypothesis is 100%. However, Sober's claim runs into the above firing squad counterexample. In an excellent article, Jonathan Weisberg considers Sober's major responses to the firing squad analogy, arguing that these responses can be dealt with via minor variations of the analogy. As he concludes, "the moral we've learned from firing squad scenarios is that we oughtn't treat our existence as an 'inevitable' background assumption, much less the fact that we'll be around to check for fine-tuning. Our existence is not inevitable, and can thus be evidentially informative."[21]

Instead of going more deeply into this debate, here I want to raise a further problem with Sober's claim, namely that it threatens to undermine scientific inference. As a specific example, consider the prediction to nine significant digits of quantum electrodynamics (QED) of the deviation from 2 of the gyromagnetic moment of the electron. Such a precise prediction was considered strongly to confirm the approximate truth (or empirical adequacy) of QED; and the reason seems to be that it was considered enormously improbable that QED's prediction would be correct if QED were not approximately true (or empirically adequate), but not improbable otherwise. Suppose, however, we discovered that the precise magnitude of deviation was necessary for the evolution of life in our universe. In that case, given Sober's claim, the probability of this deviation would be one whether or not QED is approximately true, since it follows from our existence. Yet, it seems clear that such confirmation should not depend on whether or not this deviation is necessary for life.

Finally, even if we accept Sober's claim and make our existence part of our background information, those "fine-tunings" of the universe mentioned at the end of our discussion of the multiverse – namely, the low entropy of the universe throughout, and the beauty and discoverability of the laws of nature – would still be surprising under the naturalistic single-universe hypothesis but not under theism, and thus provide evidence for the latter over the former.

C. Who Designed God Objection

One of the most common objections raised against the theistic explanation is that the designer of an artifact must be as complex as the artifact being designed. Translated into the language of fine-tuning, the idea is that the designer requires just as much

[20] For example, see Sober, E. (2004). The Design Argument. In Mann, W.E. (ed.), *Blackwell Guide to the Philosophy of Religion*. Malden, MA: Blackwell Publishers, pp. 117–147.

[21] Weisberg, J. (2005). Firing Squads and Fine-Tuning: Sober on the Design Argument. *British Journal for the Philosophy of Science* 56(4): 809–821.

fine-tuning as the thing being designed. Richard Dawkins, for instance, offers this as his main objection to the design argument (and more generally to belief in God); however, like most others who raise this objection, he simply asserts its main claim without offering any significant argument for it. This objection might apply to a finite designer, for arguably any finite agent will need some laws to determine the limits of its powers, knowledge, and other features. For example, if a being can create unicorns but not centaurs, something will have to specify why. Such specification will require fine-tuning. In contrast, an agent with unlimited power, knowledge, and other features, and who has no internal parts would not require any fine-tuning; there simply would be nothing to fine-tune. The God of traditional theism – defined as a being that is all-powerful, all knowing, and perfectly good, among other attributes – is postulated to be such a being. This is one reason why God has always been considered a being with minimal, if not zero, internal complexity.

IV. Conclusion

I argued that the evidence of fine-tuning should greatly increase our belief in the truth of theism over the naturalistic single-universe hypothesis – that is, the hypothesis that there is a single universe that exists as a brute fact. I then considered the major alternative explanation for the existence of a fine-tuned universe, the multiverse hypothesis, and raised several serious problems with it as an alternative explanation. Finally, I looked at several common objections to the argument and concluded that they fail to undercut it. Thus, I conclude, the evidence of cosmic fine-tuning discovered in the last fifty years or so provides strong support for theism over its major contenders.

Fine-Tuning is Not Evidence that God Exists

Elliott Sober

I. Introduction

Physicists use observations to estimate the values of the constants that figure in the laws of physics. Their theories indicate that it would be impossible for life to exist in our universe if the values of those constants differed more than a little from their actual values. The fine-tuning argument (FTA) begins with these scientific findings and ends with a conclusion about God's existence. The conclusion might take one of the following three forms:

(i) Therefore, God exists.
(ii) Therefore, God probably exists.
(iii) Therefore, the observed values of the physical constants favors the hypothesis that God set the values of those constants over the hypothesis that the constants had their values set by a mindless chance process.

If (i) means that God's existence can be deduced from the physical facts (perhaps when they are supplemented by independently plausible further assumptions), I think it is a non-starter. The second option also is not promising, since a conclusion about a posterior probability (the probability that God exists, given the physical facts) must be built on assumptions about the value of the prior probability that God exists, and it seems impossible to justify a claim about the value of that prior probability or even a claim about what its range of values is (short of the trivial and unhelpful remark that it falls between 0 and 1). That leaves option (iii).

II. The Law of Likelihood

I interpret the concept of favoring used in option (iii) in terms of the following principle:

(**Law of Likelihood**) Observation O favors hypothesis H_1 over hypothesis H_2 if and only if $Pr(O \mid H_1) > Pr(O \mid H_2)$.[22]

Notice that this law says nothing about the probabilities of hypotheses – for example, about whether $Pr(H_1 \mid O) > Pr(H_2 \mid O)$.

A simple example conveys the intuitive appeal of this principle, though it is no substitute for a careful consideration of the law's theoretical justification. Consider two hypotheses about the composition of an urn that contains 10,000 balls:

(X) 80% of the balls in the urn are green and 20% are red.
(Y) 10% of the balls in the urn are green and 90% are red.

To evaluate these hypotheses, you close your eyes, draw 100 balls at random from the urn (without replacement), and then observe that 90 of them are green and 10 are red. This observation favors X over Y, though of course it does not prove that one is true and the other is false.

When the FTA is given a likelihood formulation, its conclusion will be something like the following:

(*) Pr(the value of physical constant x is in window W | God set the values of the constants & W is the narrow range of values for x that permit life to exist) > Pr(the value of physical constant x is in window W | a mindless chance process set the value of x & W is the narrow range of values for x that permit life to exist).

Here I describe the FTA in terms of what it says about the value of a single physical constant, but that is a simplification. The likelihoods of God and Chance should

[22] Hacking, I. (1965). *The Logic of Statistical Inference*. Cambridge: Cambridge University Press.

be compared, given the values of *all* the physical constants, not just one of them. This complication will not affect the points I will endeavor to make.

The probabilities under consideration in the FTA describe facts about the world; they do not, in the first instance, describe facts about the beliefs an agent has, or should have, about the world. In just the same way, it is an objective (mind-independent) fact in the urn example that the observations are more probable under hypothesis X than they are under hypothesis Y if there is random sampling. This is not to deny that rational agents may want their degrees of belief to mirror various objective facts.

The (*) inequality looks like it must be true, since the left-hand probability seems to be big, and the right-hand probability seems to be small. However, there are two questions. The first concerns the God hypothesis. If God is life-loving, maybe it follows that the left-hand likelihood is large, but *is* God life-loving? This might be doubted, as there is so little life in the universe. Here one finds a parallel with the problem of evil. The problem of evil is not to explain why there is some evil rather than none at all. Rather, the problem is to explain why there is so much evil rather than less. The present question is: why is there so little life rather than more?[23] The other question about (*) concerns the Chance hypothesis. McGrew, McGrew, and Vestrup argue that this hypothesis lacks a well-defined and non-arbitrary likelihood;[24] Collins disagrees.[25] Space limitations prevent me from discussing these questions. In what follows, I will assume for the sake of argument that the (*) inequality is true. This may suggest, erroneously, that I have no further objections to make.

III. An Improved Formulation of the Law of Likelihood

There is a disconnect between the law of likelihood as described so far and the urn example. The law mentions the hypotheses and the observations, but says nothing about how those observations were obtained. Yet, it was part of my story about the urn that you drew balls at random and did not return balls to the urn after you had examined them. That was no idle detail, since the same observations might not favor X over Y if a different sampling scheme had been used. For example, suppose that, unbeknownst to you, your assistant in another room reached into the urn and deliberately assembled 90 green balls and 10 red. The assistant then brings you those 100 balls to inspect. You might naively think that your observations favor hypothesis X over hypothesis Y, but

[23] The response to the problem of evil called "skeptical theism" asserts that we cannot understand God's reasons for allowing so much evil to exist. Should theists who find this plausible also be loath to endorse design arguments for the existence of God, including the FTA? See Sober, E. (2003). The Design Argument. In N. Manson (ed.), *God and Design: The Teleological Argument and Modern Science*. London: Routledge; and Sober, E. (2019). *The Design Argument: An Element*. Cambridge: Cambridge University Press.

[24] McGrew, T., McGrew, L., and Vestrup, E. (2001). Probabilities and the Fine-Tuning Argument: A Skeptical View. *Mind* 110: 1027–1037.

[25] Collins, R. (2009). The Teleological Argument: An Exploration of the Fine-Tuning of the Universe. In W.L. Craig and J.P. Moreland (ed.), *The Blackwell Companion to Natural Theology*. Oxford: Wiley Blackwell, pp. 202–281.

you would be wrong. To take this point into account, I will reformulate the law of likelihood so that the role of the process of observation is made explicit:

(**Improved Law of Likelihood**) Given that P is true, where P is a proposition describing the process by which the observation was obtained, observation O favors hypothesis H_1 over hypothesis H_2 if and only if $Pr(O \mid H_1\&P) > Pr(O \mid H_2\&P)$.[26,27]

Proposition (*), the preliminary formulation of the conclusion of the fine-tuning argument, thus needs to be reconfigured so that it takes account of the process by which we observe the value of physical constant x.

IV. Eddington's Fishing Net

Eddington tells the following story.[28] You use a net to fish in a lake. You observe that all the fish in the net are more than 10 inches long (call this proposition O). You then consider two hypotheses about the lake's composition:

(H_1) All the fish in the lake are more than 10 inches long.
(H_2) Half the fish in the lake are more than 10 inches long.

You apply the law of likelihood and conclude that your observation favors H_1 over H_2. Then, turning your attention to the net, you notice that it has holes that are big enough to allow fish that are less than 10 inches long to easily escape. Call this proposition B, for big. This discovery leads you to revise your assessment of the evidence. You reason that the observation does not favor H_1 over H_2, since $Pr(O \mid H_1\&B) = Pr(O \mid H_2\&B)$. Given the construction of the net, the probability of proposition O is the same, regardless of whether H_1 or H_2 is true. The net you used induced an *observational selection effect* (an OSE). Had you used a net with small holes, you could have avoided the problem. The OSE introduced a *bias* into the observational process.[29]

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[26] The vanilla version of the law of likelihood is not wrong, since assumptions about the process of observation can be "baked into" the probability function Pr(-). Still, for the sake of explicitness, we should note the baking, perhaps by writing $Pr_p(O \mid H_1) > Pr_p(O \mid H_2)$.

[27] Kotzen prefers a different reformulation of the law of likelihood, one that also takes account of both O (the observation) and P (the process by which observation O was obtained):

(LP**) The fact that O was collected through a process of observation characterized by P is evidence for H_1 over H_2 if $Pr(O\& P \mid H_1) > P(O\&P \mid H_2)$.

See Kotzen, M. (2012). Selection Biases in Likelihood Arguments. *British Journal for the Philosophy of Science* 63: 825–839. I do not think LP** is better than my improvement. First, his addresses a different question. Kotzen is interested in whether O&P provides discriminating evidence. I am interested in whether O provides discriminating evidence, given that P is true. Second, LP** does not indicate that you need to consider the process by which O&P was obtained.

[28] Eddington, A. (1939). *The Philosophy of Physical Science*. Cambridge: Cambridge University Press.

[29] Sober, E. (2009). Absence of Evidence and Evidence of Absence: Evidential Transitivity in Connection with Fossils, Fishing, Fine-Tuning, and Firing Squads. *Philosophical Studies* 143: 63–90. Also see Sober (2003). op. cit.

The bias is there whether you know it or not. If you are unsure about what sort of net you used, you might assign a degree of belief p to your having used a net with big holes and a degree of belief 1 – p to your having used a net with small holes (S). You then would conclude that your evidence favors H_1 over H_2, since

$$\Pr(O \mid H_1) = \Pr(O \mid H_1 \& B)p + \Pr(O \mid H_1 \& S)(1 - p) >$$
$$\Pr(O \mid H_2) = \Pr(O \mid H_2 \& B)p + \Pr(O \mid H_2 \& S)(1 - p).^{30}$$

This is how a subjective Bayesian would address the problem, but the concept of an OSE has a different import. The physical facts about the net you *actually* used induce an OSE. Your *uncertainty* about which net you used is irrelevant.[31]

The fine-tuning argument asserts a likelihood inequality, but once you take account of the process of observation, you see that the relevant likelihoods are equal. I will use the phrase "the constants are right" to mean that the constants fall in the narrow window of values that permit life to exist in our universe. The FTA asserts the following inequality:

Pr(the constants are right | God set the values of the constants) >
Pr(the constants are right | the values of the constants were set by a mindless chance process).

The replacement I have in mind is this:

(**) Pr(you observe that the constants are right | God set the values of the constants & P) = Pr(you observe that the constants are right | the values of the constants were set by a mindless chance process & P).

P describes facts about your process of observation. The (**) equality uses the improved law of likelihood; it implements the *weak anthropic principle*, which Carter characterized by saying that "what we can expect to observe must be restricted by the conditions necessary for our presence as observers."[32]

Notice that the evidence statement in (**) (the statement to the left of the conditional probability sign) is not the claim that the constants are right; rather, it is the logically stronger claim that *you observe* that the constants are right. Here I abide

[30] This inequality is true because $0 < p < 1$, $\Pr(O \mid H_1\&B) = \Pr(O \mid H_2\&B)$, and $\Pr(O \mid H_1\&S) > \Pr(O \mid H_2\&S)$.

[31] Monton treats the question of whether the FTA is subject to an OSE within the context of subjective Bayesianism. See Monton, B. (2006). God, Fine-Tuning, and the Problem of Old Evidence. *British Journal for the Philosophy of Science* 57: 405–424.

[32] Carter, B. (1974). Large Number Coincidences and the Anthropic Principle in Cosmology. In M.S. Longair (ed.), *Confrontation of Cosmological Theories with Observational Data*. Dordrecht: Reidel, pp. 291–298. For discussion of other anthropic principles, see Earman, J. (1987). The SAP Also Rises: A Critical Examination of the Anthropic Principle. *American Philosophical Quarterly* 24: 307–317.

by the principle of total evidence.[33] Sometimes it makes no difference whether you use a weaker or a stronger formulation; for example, in the urn problem, it does not matter whether you use "there were 90 green and 10 red balls in the sample" or "you observe that there are 90 green and 10 red balls in the sample." Still, OSEs often need to be analyzed by using the stronger formulation of the evidence.

How should the process of observation in (**) be described? One possibility is to cite the fact that you are alive. Given that you are alive, the probability of your observing that the constants are right is the same, regardless of whether God or Chance is true. Nothing would change if you were somehow unsure about whether you are alive. Your study of philosophy may have led you to wonder whether you are a living thing, a lifeless robot, or an incorporeal being.[34] Your uncertainty does not matter. The fact is that you are alive. You drag that fact behind you like a net.

V. Leslie's Firing Squad

The similarity of fishing and fine-tuning may seem compelling, leading to the conclusion that the FTA is undermined by an OSE, but Leslie describes a clever example that seems to throw doubt on the claimed similarity.[35] You face a firing squad. They fire, and shortly thereafter you are surprised to find that you are alive and unscathed. You conclude that your being alive is evidence that favors the first of the following hypotheses over the second:

(H_3) The firing squad decided to spare you.

(H_4) The firing squad decided to kill you.

It seems obvious that there is a likelihood inequality here, namely

$$\Pr(\text{you are alive} \mid H_3) > \Pr(\text{you are alive} \mid H_4).$$

[33] I will explain this principle via the urn example. Recall that you were testing two hypotheses about the composition of the urn and used the observation that 90 of the balls sampled were green and 10 were red. This observation favors hypothesis X over hypothesis Y. But now consider a different true description of what you observed – each of the 100 balls sampled was either green or red (O_w). This description fails to discriminate between the two hypotheses, since

$$\Pr(O_w \mid X) = \Pr(O_w \mid Y).$$

This equality, though true, is not relevant to the problem at hand. The following principle explains why:

(**Principle of Total Evidence**) In evaluating whether your observations favor one hypothesis over another, use the logically strongest true description of your observations on which each of the competing hypotheses confers a probability.

This is a pragmatic principle – it is about which observation statements to *use*, not which are *true*.

[34] Weisberg, J. (2005). Firing Squads and Fine-Tuning: Sober on the Design Argument. *British Journal for the Philosophy of Science* 56 (4): 809–821.

[35] Leslie, J. (1989). *Universes*. London: Routledge.

It also seems obvious that this inequality should not be dislodged by conditionalizing on your being alive:

$$\Pr(\text{you are alive} \mid H_3 \,\&\, \text{you are alive}) = \Pr(\text{you are alive} \mid H_4 \,\&\, \text{you are alive}).$$

This equality is true but irrelevant. There is no OSE here. You now can see why it is a delicate matter to decide which propositions to use in describing the process of observation in (**). Should fine-tuning be modelled on Eddington's fishing net or on Leslie's firing squad?

VI. Separating Source, Observational Process, and Observational Outcome

To analyze this problem, it helps to use time indices for the different propositions that might characterize a Source at t_1, an Observational Outcome at t_3, and the Observational Process that occurs in between, at t_2.[36] The three times can be temporal intervals; they need not be temporal instants. Here is how I propose to separate these three items in the three examples:

Fishing: The content of the lake is fixed at t_1 (and remains the same thereafter). At t_2 you put your net in the lake. At t_3 you observe that all the fish in the net are longer than 10 inches.

Fine-tuning: God or Chance sets the values of the constants at t_1 (and the values remain the same thereafter). At t_2 you are alive. At t_3 you observe that the constants are right.

Firing-squad: The firing squad decides at t_1 what it will do at t_2 (and the decision remains in place thereafter). You are alive when the squad fires at t_2. At t_3 you observe that you are alive.

In each example, what happens at one time can affect the probability of what happens next, but does not logically necessitate what happens next.

An additional constraint is needed concerning how the process mediating the connection of source to observation should be understood. It can be gleaned from the children's game of Telephone. Ruby whispers a word to Ezra, then Ezra whispers to Lilah what he thinks Ruby said. Lilah then whispers to a fourth child, and so on. The game ends, typically in giggles, when the last child in the chain reports what he or she heard and the first child then tells everyone what she said. For simplicity, I will focus on the three children mentioned, and I will suppose that each player must choose between "apple" and "orange" as the word that he or she will say. Lilah's hearing "apple" is evidence concerning what Ruby said, since

$\Pr(\text{Lilah hears "apple" at } t_3 \mid \text{Ruby said "apple" at } t_1) >$
$\Pr(\text{Lilah hears "apple" at } t_3 \mid \text{Ruby said "orange" at } t_1).$

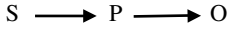
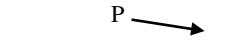
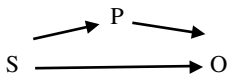
[36] Sober (2009). op. cit.

However, consider the following equality:

Pr(Lilah hears "apple" at t_3 | Ruby said "apple" at t_1 & Ezra said "apple" at t_2) =
Pr(Lilah hears "apple" at t_3 | Ruby said "orange" at t_1 & Ezra said "apple" at t_2),

This statement is true, but it is irrelevant to deciding whether Lilah's impression at t_3 is evidence concerning what Ruby said at t_1. Ezra at t_2 is part of the process by which Ruby at t_1 affects Lilah at t_3, but you should not consider what Ezra said in deciding whether Lilah's state is evidence concerning what Ruby said.

Possible causal relationships among source (S), process (P), and observation (O)

	t_1 t_2 t_3
I	
II	
III	

The game of Telephone suggests a general point, represented in the accompanying figure. In pattern I, S influences O only by way of the intermediate process P. The Telephone game conforms to pattern I. In pattern II, S and P are probabilistically independent causes of O. In III, there are two pathways from S to O, one of which passes through P, and S and P are probabilistically dependent. In considering whether the relationship of Source, Process, and Observation exemplifies an OSE, it is patterns II and III that are relevant. One does not demonstrate that an OSE is in place by showing that the variables exhibit pattern I. Fishing, fine-tuning, and the firing squad each exhibit pattern II or pattern III.

VII. The Three Examples

Each of the three accompanying tables describes the probability of an observational outcome conditional on different hypotheses about the source and different propositions about the observational process. The three tables have two columns, but only the left-hand column is relevant to the likelihood analysis, since I assume that the propositions that label that column are true and the ones that label the right-hand column are false. I include those right-hand columns for a reason I will soon explain.

Fishing: Pr(you observe at t_3 that the net contains fish all more than 10 inches long | —)			
		Process	
		You put a net with big holes in the lake at t_2.	You put a net with small holes in the lake at t_2.
Hypotheses	All the fish in the lake are more than 10 inches long at t_1.	=	
	Half the fish in the lake are more than 10 inches long at t_1.	>	<

Fine-Tuning: Pr(you observe at t_3 that the constants are right | —)			
		Process	
		You are alive at t_2.	You are not alive at t_2.
Hypotheses	God sets the constants at t_1.	>	0
	Chance sets the constants at t_1.	>	0

Firing Squad: Pr(you observe at t_3 that you are alive | —)			
		Process	
		You are alive at t_2.	You are not alive at t_2.
Hypotheses	Squad decides at t_1 to spare you when it fires just after t_2.	>	0
	Squad decides at t_1 to kill you when it fires just after t_2.	>	0

With respect to the left-hand columns in these tables, fine-tuning resembles fishing, not the firing squad. Here are the likelihood equalities that pertain to the first two, and the likelihood inequality that characterizes the third:[37]

Fishing: Pr[You observe at t_3 that all the fish in the net are more than 10 inches long | at t_1 all the fish in the lake are more than 10 inches long (and that remains so) & at t_2 you put a net with big holes in the lake] = Pr[You observe at t_3 that all the fish in the net are more than 10 inches long | at t_1 half the fish in the lake are more than 10 inches long (and that remains so) & at t_2 you put a net with big holes in the lake].

[37] Sober (2009). op. cit., pp. 83–86.

Fine-tuning: Pr[You observe at t_3 that the constants are right | at t_1 the values of the constants are set by an intelligent designer (and the values do not change) & at t_2 you are alive]	=	Pr[You observe at t_3 that the constants are right | at t_1 the values of the constants are set by a mindless chance process (and the values do not change) & at t_2 you are alive].
Firing squad: Pr[You observe at t_3 that you are alive | at t_1 the firing squad decides to spare you when it fires at t_2 (and their decision does not change) & at t_2 you are alive and the squad fires]	>	Pr[You observe at t_3 that you are alive | at t_1 the firing squad decides to kill you when it fires at t_2 (and their decision does not change) & at t_2 you are alive and the squad fires].

Why are fine-tuning and the firing squad different? In fine-tuning, your being alive at t_2 guarantees that the constants are right at t_1, t_2, and t_3, so the probability of your observing at t_3 that the constants are right is the same regardless of whether it was God or Chance that set their values. However, your being alive at t_2 in the firing squad example leaves open what the firing squad decided at t_1, with the result that your observing at t_3 that you are alive provides evidence about what the squad decided at t_1.[38] The fact that you are alive at t_2 induces an OSE in one example but not in the other.

Fine-tuning and the firing squad are the same with respect to their right-hand columns. This may account for why fine-tuning and the firing squad seem similar. However, as mentioned, that similarity is epistemically irrelevant. My exclusive focus on left-hand columns in these three examples reflects my earlier point that whether you are the victim of an OSE is an objective fact about your process of observation; your uncertainty about the character of that process is not relevant.

VIII. The Multiverse Hypothesis

In discussing the FTA, I have compared two hypotheses (God and Chance), but there is a third hypothesis that has been considered. The God and Chance hypotheses both assume that there is a single universe. The multiverse hypothesis postulates that there are infinitely many universes that are causally unconnected with each other, and that the values of the constants in each universe are set independently by a single mindless chance process that can have different outcomes for different universes. This set-up seems to make it overwhelmingly probable that some universe or other will have constants that are right for life. If so, it follows that

Pr(some universe has constants that are right | Multiverse) =
Pr(some universe has constants that are right | God chose the constants in our Universe),

which entails that the observation does not favor God over the Multiverse. This equality might be true, but, if it is, so is the following inequality:

[38] Ibid., p. 86.

Pr(our universe has constants that are right | Multiverse) <
Pr(our universe has constants that are right | God chose the constants in our Universe).

The principle of total evidence entails that you should focus on *our* universe having constants that are right, not on *some* universe having constants that are right.[39] However, there is an OSE at work when we observe that the constants in our universe are right, so the above inequality gets replaced by an equality, though it is not the equality displayed in the present paragraph.[40]

IX. Concluding Comments

The FTA begins with an observation and a counterfactual. The observation is that the constants in the laws of physics L have values that fall in narrow windows. The counterfactual is that life would not be possible in our universe if the values of those constants were to fall outside those windows. The FTA claims that the observation favors God over Chance, given that the counterfactual is true. This argument is vitiated by an OSE, but this deficiency leaves it open that some other observation might discriminate between the two hypotheses without falling prey to the same objection.[41]

There is a counterfactual, slightly different from the one just stated, to consider here. Setting aside L, consider the more abstract proposition that the laws of nature (whatever they are) would not permit life to exist, if the constants in them were to have values that differed more than a little from their actual values. Does this counterfactual favor God over Chance? The suggestion that it does is not undermined by an OSE.[42] Even so, it is mysterious why this counterfactual should have a higher probability under the God hypothesis than it has under the hypothesis of Chance. Could God not have had life in the universe by choosing laws that do not need to be fine-tuned?[43] If so, why would God want the laws of nature to require fine-tuning for life to exist?[44]

Just as the FTA is undermined by an OSE, the same is true when other observations are considered. For example, if the observation is that consciousness exists, or that bodies exist, an OSE is induced by the fact that the observers making those observations are conscious and embodied.

I have taken pains to sort out the similarities and differences that connect the fine-tuning argument to Eddington's fishing net and to Leslie's firing squad, but I have

[39] For discussion, see Hacking, I. (1987). The Inverse Gambler's Fallacy: The Argument from Design and the Anthropic Principle Applied to Wheeler Universes. *Mind* 96: 331–340; White, R. (2000). Fine-Tuning and Multiple Universes. *Nous* 34: 260–276; and Sober, E. (2003). The Design Argument. In N. Manson (ed.), *God and Design: The Teleological Argument and Modern Science*. London: Routledge.

[40] Collins (2009). op. cit., argues that the multiverse hypothesis is inferior to the God hypothesis.

[41] See Sober (2009). op. cit.

[42] For discussion, see Roberts, J. (2012). Fine-Tuning and the Infrared Bulls-Eye. *Philosophical Studies* 160: 287–303; and McGrew, L. (2016). Four (or so) New Fine-Tuning Arguments. *European Journal for Philosophy of Religion* 8: 85–106.

[43] Weisberg, J. (2010). A Note on Design: What's Fine-Tuning Got to Do with It? *Analysis* 70 (3): 431–438.

[44] This question resembles a query that needs to be considered in connection with design arguments concerning the adaptations we observe (e.g. the vertebrate eye). To tell whether the features of the eye favor God over Chance, we need information about what God's goals would be in providing vertebrates with adaptive features. See Sober (2003).

tried to do more than this. I have tried to develop a general framework for thinking about OSEs. It will not surprise me if that framework needs to be fine-tuned.

I have formulated the FTA as a likelihood argument, and have argued that it faces a serious problem, but maybe the likelihood approach was wrong from the start. This possibility is worth considering.[45] It is distinct from the suggestion that the likelihood approach is *incomplete*; Bayesians think this is so, and want to add prior probabilities to the mix. In doing so, they are not rejecting the relevance of likelihood comparisons.[46]

Reply to Sober

Robin Collins

Sober's main claim is that the observer-selection effect (OSE) nullifies the likelihood version of the FTA, for example, as presented in my main essay. A major problem with Sober's argument is that he misses what is rendered improbable by the fine-tuning evidence, namely the *existence* of embodied conscious agents (ECAs). Instead, he focuses on the probability that, given you exist and make the relevant observations about the values of the constants, you will observe that they are set just right for life. To justify his formulation of the argument, he invokes what he calls the improved law of likelihood: namely, "Given that P is true, where P is a proposition describing the process by which the observation was obtained, observation O favors hypothesis H_1 over hypothesis H_2 if and only if $\Pr(O \mid H_1\&P) > \Pr(O \mid H_2\&P)$." He motivates this improved law by giving an example of picking balls from an urn where the process P is not random.

Applied to the fine-tuning, we can simplify P to be the proposition that you exist (and attempt to observe the values of the constants), and O being the claim that you observe the constants to fall into a life-permitting window, and your background information k′ being that this window is relatively very small relative to an appropriately chosen comparison range. Since you would not exist unless O were true, P entails O and hence $\Pr(O \mid H_1\&P\ \&\ k') = \Pr(O \mid H_2\&P\ \&\ k')$, no matter what H_1 and H_2 are. This equality, however, is irrelevant to the fine-tuning argument, since the fine-tuning concerns your existence, not whether *given you exist*, you will observe the universe you are in to be fine-tuned for life. Since the process, P, by which you obtained the evidence O entails your existence, in the case of the fine-tuning argument, P cannot be included in the conditional (the part after the | symbol).

Put symbolically, the fine-tuning argument relies on the claim that P(ECAs exist | T & k′) ≫ P(ECAs exist | NSU & k′), where ≫ is much greater than, k′ is one's

[45] See White, R. (2015). The Argument from Cosmological Fine-Tuning. In G. Rosen, A. Byrne, J. Cohen and S. Shiffrin (ed.), *The Norton Introduction to Philosophy*. New York: Norton, pp. 45–50. White thinks the FTA assumes that it needs to be explained why life is possible in our universe. The premise is wrong, he says, if the multiverse hypothesis is true, since that hypothesis removes the need to explain why life is possible in our universe since it says that it is overwhelmingly probable that life should exist in some universe or other. The likelihood assessment I have developed here says nothing about what needs to be *explained*. The question is whether a given observation is *evidence* that discriminates among hypotheses.

[46] My thanks to Dylan Beschoner, Jordan Ellenberg, Lydia McGrew, Brian Mcloone, Steven Nadler, Shannon Nolen, Jim Paulson, William Roche, Larry Shapiro, Mike Steel, Reuben Stern, Michael Titelbaum, Olav Vassend, and Peter Vranas for useful discussion.

background information with any information about whether the constants are right for ECAs subtracted out.[47] Presumably, Sober thinks these probabilities are illegitimate because, presumably, he assumes that one must include P in one's background information. However, I do not see where he argues for that, at least not in the essay at hand. Further, it seems wrong in our ordinary use of probability. Consider, for example, the common claim that if the dinosaurs were not wiped out, it is very unlikely that we would exist: that is, the claim that P(we exist | dinosaurs were not wiped out) is close to zero. If we required that we always include in our background information P, then this probability would become P(we exist | dinosaurs were not wiped out & P) = 1, since P entails our existence.

In the section of his essay titled "The Three Examples," Sober addresses Leslie's firing-squad analogy and then attempts to explain why it is relevantly different from the case of fine-tuning:

> In fine-tuning, your being alive at t_2 guarantees that the constants are right at t_1, t_2, and t_3, so the probability of your observing at t_3 that the constants are right is the same regardless of whether it was God or Chance that set their values. However, your being alive at t_2 in the firing squad example leaves open what the firing squad decided at t_1, with the result that your observing at t_3 that you are alive provides evidence about what the squad decided at t_1. The fact that you are alive at t_2 induces an OSE in the one case but not in the other.

So, from the above quotation, "… your being alive at t_2 in the firing squad example leaves open what the firing squad decided at t_1," and thus, "… the result that your observing at t_3 that you are alive provides evidence about what the squad decided at t_1." In response, note that an exactly parallel line of reasoning can be constructed in the case of fine-tuning: your being alive at t_2 leaves open whether God or chance decided the values of the constants at t_1, and hence your being alive at t_3 provides evidence about whether God or chance set the values at t_1. Further, one can also draw an almost exact parallel in the case of firing squad with Sober's first sentence in the quotation above. Namely:

> In firing squad, your being alive at t_3 guarantees that at t_2 bullets were fired in such a way that they would miss, and so the probability of your observing at t_3 that the bullets missed is the same regardless of whether the firing squad at t_1 intentionally aimed them to miss *or* whether at t_1 they aimed them to kill but by chance they were fired (at t_2) in such a way that they would miss.

The only potentially relevant difference between fine-tuning and firing squad that I can see is that in the firing squad case there is a time t_2 when you were alive but at which time it was still an open question whether the bullets were fired in such a way as to hit or miss. From what Sober says, however, I fail to see why this is relevant. But even if it is, there is a variation of the firing squad example due to Jonathan Weisberg in which this difference vanishes: namely, one in which your mother was before the firing squad.[48]

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[47] I spell this out in detail in Collins, R. (2009). The Teleological Argument: An Exploration of the Fine-Tuning of the Universe. In W.L. Craig and J.P. Moreland (ed.), *The Blackwell Companion to Natural Theology*. Oxford: Wiley Blackwell, pp. 202–81.

[48] Weisberg, J. (2005). Firing Squads and Fine Tuning: Sober on the Design Argument. *The British Journal for the Philosophy of Science* 56 (4): 809–821. Especially footnote 9, p. 820.

Assuming you would not have existed without your mother giving birth to you, there would be no time before you existed in which it was open whether the bullets were fired to hit or to miss.

Reply to Collins

Elliott Sober

I. What is the Fine-Tuning Argument's Conclusion?

I take the conclusion of the fine-tuning argument (FTA) to be:

(1) $\Pr(L \mid G \& Cf) > \Pr(L \mid Ch \& Cf)$.
L is the proposition that the physical constants have values that are life-permitting, G is the hypothesis that God exists and set the values of the physical constants, Ch is the hypothesis that a mindless chance process set the values of the constants, and Cf is the counterfactual that there would be no life if the constants had values that differ more than a little from their actual values.

Collins sometimes formulates the FTA differently, taking the observation to be that conscious embodied agents exist. To me this is a modest difference, since I think that all the problems that attach to my version also attach to his. A more substantial difference arises when Collins takes the FTA to be arguing that Cf is evidence that favors God over Chance. According to the law of likelihood, this means that

(2) $\Pr(Cf \mid G) > \Pr(Cf \mid Ch)$.
Notice that the God and Chance hypotheses in (1) and (2) are different. In (1) the laws are taken as given and G and Ch assign probabilities to the values of the *constants* in those laws being life-permitting. In (2), G and Ch assign probabilities to a single proposition about the *laws* themselves. Proposition (2) is dubious. Why would a life-loving God want Cf to be true? Why did God not choose different laws of nature that would allow life to exist across a broader range of possible values for their constants?[49] Collins argues that Cf is evidence for God over Chance by describing a lottery example; he thinks that what is true of the lottery shows that Cf is evidence, and that there is no need to evaluate the probabilities in (2) (personal communication). I do not think this lottery example does the trick, but I will not pursue the point here; in addition, Collins's discussion in his paper of why Cf is evidence sounds like it involves a likelihood comparison.

II. Epistemic Probabilities

Collins thinks the probabilities in the FTA are "epistemic," meaning that Pr(A | B) represents the degree of support that B *of itself* confers on A, by which Collins means that you should think about the value of this conditional probability by bracketing

[49] Weisberg, J. (2010). A Note on Design: What's Fine-Tuning Got to Do with It? *Analysis* 70 (3): 431–438.

all other sources of support or warrant for A or for not A (and any undercutting defeaters of A). I think this entails that when you evaluate $Pr(A \mid B)$, you should not consider the value of $Pr(A \mid B\&C)$, if C is itself evidentially relevant to A in the sense that $Pr(A \mid C) \neq Pr(A)$.

This entailment has disastrous consequences when it is applied to Eddington's fishing example. Suppose O is the observation that all the fish in your net are more than 10 inches long, H_1 is the hypothesis that all the fish in the lake are more than 10 inches long, and H_2 is the hypothesis that only 50% of the fish in the lake are more than 10 inches long. Is O evidence favoring H_1 over H_2? Collins and I agree that it is not, because once you take the process of observation into account, you obtain the following equality:

$$Pr(O \mid H_1 \& B) = Pr(O \mid H_2 \& B).$$

Here B is the proposition that you used a net with big holes. Unfortunately, Collins's own conception of epistemic probability prohibits this analysis if there is some chance that there are small fish in the lake and there was some chance that you might have used a net with small holes, since then $Pr(O \mid B) \neq Pr(O)$.

Collins's conception of epistemic probabilities requires a firm distinction between the evidence that bears on O and the process of observation used to obtain O. As noted, he wants to take the process of observation into account but wishes to bracket other sources of evidence that bear on O. The problem is that the process we use to make our observations is sometimes also evidence pertinent to whether O is true.

III. The Restricted Principle of Indifference

A factory makes cubes and they are always between 1 and 5 inches on an edge.[50] What is the probability that a cube, randomly selected from the factory output, will have an edge that is between 1 and 3 inches? If you knew more about the manufacturing process you could answer this question, but what can you say based just on the information I have supplied? The principle of indifference has an answer. You should assume that the probability is ½, because you have no reason to think that the edge's being between 1 and 3 inches differs in probability from its being between 3 and 5 inches. This probability distribution is flat, meaning that the two possible outcomes are assigned the same probability. The principle of indifference is a device for deriving probabilities from ignorance.

There is a problem. If the cubes are always between 1 and 5 inches on an edge, then they are always between 1 and 125 cubic inches in volume. The principle of indifference concludes that the probability of the cube's having a volume between 1 and 63 cubic inches is ½. This entails that the probability is ½ that its edges will be between 1 and $\sqrt[3]{63}$ (which is approximately 4) inches.

The principle of indifference yields contradictory advice if it is applied to both the cube's edge length and its volume. The principle therefore needs to be restricted, and

[50] The example is from Van Fraassen, B. (1989) *Laws and Symmetries*. Oxford: Oxford University Press, p. 303.

the restriction needs to be justified. Collins's *restricted* principle of indifference is supposed to solve this problem. It says that you should use the most natural descriptor, where naturalness means that the descriptor represents a real physical quantity *or* that it is used in the simplest formulation of the fundamental laws of physics. Length and volume are both real physical quantities and Collins's "or" means that both are natural, so the contradiction that arises in the cube problem is not evaded. Does the simplest formulation of the fundamental laws of physics mention length but not volume, or volume but not length? Even if the answer is *yes*, I do not see why simplicity justifies assigning a flat probability distribution to one of them but not to the other. Ockham's razor, properly understood, is not up to the job.[51] Collins, in his footnote 10, recommends averaging or blending distributions that are each based on natural descriptors, thus obtaining a single distribution. Perhaps this guarantees a unique solution, but the problem remains of justifying why we should opt for the single distribution thus constructed.

The situation gets worse when the principle of indifference, restricted or not, confronts the FTA. It is clear enough that the left-hand probability in (1) is big, assuming that God is life-loving, omnipotent, and omniscient. The problem arises on the right-hand side. What probability distribution for possible value ranges of the physical constants is the Chance hypothesis using?

In the cube problem, the value range is finite (4 inches). The Chance hypothesis in the FTA presents a harder problem, since the range of logically possible values for many physical constants is $[0,\infty]$ – all finite non-negative numbers are logically possible. If sense cannot be made of the right-hand side of (1), then no good argument can reach the conclusion that (1) is true, and the FTA goes up in smoke. This is the conclusion that McGrew et al. draw, because they demand that the Chance distribution be nonarbitrary and normalizable, where the latter means that the probabilities assigned to the exclusive and exhaustive possible value ranges for a physical constant must sum to one.[52] Collins earlier rejected the requirement of normalizability,[53] but he does not discuss that in the present paper. Rather, he mentions his concept of an *epistemically illuminated region*. The idea is to ignore the fact that there is an infinitely wide window of logically possible values for a physical constant and focus instead on the fact that there is a finitely wide window encompassing all and only the values about which we can be reasonably confident whether or not they permit conscious embodied agents to exist. I do not understand why that window is not infinitely wide, nor why this suggestion makes precise what the Chance hypothesis says. That hypothesis, as I understand it, does not assign zero probability to the values of constants about which we are unable to say whether they are compatible with the existence of conscious embodied agents.[54]

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[51] Sober, E. (2015). *Ockham's Razors: A User's Manual*. Cambridge: Cambridge University Press.

[52] McGrew, T., McGrew, L., and Vestrup, E. (2001). Probabilities and the Fine-Tuning Argument: a Skeptical View. *Mind* 110: 1027–1037.

[53] Collins, R. (2009). op. cit., pp. 249 ff.

[54] For more on this mathematical aspect of the FTA, see Sober, E. (2019). *The Design Argument*. Cambridge: Cambridge University Press, §5.2.

IV. Observation Selection Effects

Collins takes Leslie's firing squad example to undermine the relevance of OSEs to the FTA, partly because he endorses Weisberg's[55] criticisms of my 2003 paper.[56] A more recent paper of mine provided a different treatment of OSEs,[57] and so does my essay in the present volume.

Collins thinks my contention that an OSE refutes the FTA has a disastrous consequence – it threatens to undermine scientific inference. He argues for this by describing a single hypothetical example. Suppose QED (quantum electrodynamics) made a prediction (P) whose correctness was widely taken to strongly support the theory. Collins asks the reader to imagine that P's being true turns out to be necessary for life to exist. He claims that this turn of events would not prevent the successful prediction from strongly confirming QED.

If Collins's example is to address the likelihood formulation I use for conceptualizing OSEs, it needs to contrast QED with a specific alternative hypothesis. To do this, let us suppose that QED competes with theory X, and that X says that P is very improbable. If P must be true for life to exist in the universe, then

Pr(you observe at t_3 that P is true | QED is true at t_1 and subsequently & you are alive at t_2) =
Pr(you observe at t_3 that P is true | X is true at t_1 and subsequently & you are alive at t_2).

Collins thinks that recognizing the OSE in this example would open the flood gates – that it would undermine legitimate scientific inference *generally.* I think this dire forecast is greatly exaggerated. Many scientific inferences are unaffected by OSEs;[58] those that are affected face a problem that should not be swept under the rug.[59]

Suggestions for Further Reading

In these essays, the authors cite many articles that together give a thorough picture of the contemporary discussion of fine-tuning. A few longer works that are cited but bear repeating include a classic statement of the fine-tuning argument by John Leslie called *Universes* (1989), and Martin Reese's *Just Six Numbers: The Deep Forces that Shape the*

[55] Weisberg, J. (2005). Firing Squads and Fine-Tuning: Sober on the Design Argument. *British Society for the Philosophy of Science* 56 (4): 809–821.

[56] Sober, E. (2003). The Design Argument. In N. Manson (ed.), *God and Design: The Teleological Argument and Modern Science*. London: Routledge pp. 25–53.

[57] Sober, E. (2009). Absence of Evidence and Evidence of Absence: Evidential Transitivity in Connection with Fossils, Fishing, Fine-Tuning, and Firing Squads. *Philosophical Studies* 143: 63–90.

[58] Consider two of the examples from my paper – random sampling from an urn and fishing with a net that has small holes.

[59] I thank Joel Ballivian, Robin Collins, and William Roche for useful comments on an earlier version of this paper.

Universe (2000) which indicates how finely tuned the universe appears to be. Additional resources include the collection *God and Design: The Teleological Argument and Modern Science* (2003) edited by Neil Manson; *God or Blind Nature: Philosophers Debate the Evidence* (2007–2008), an online book available at the website www.infidels.org; and the entry on "Fine-tuning" (2017) in the *Stanford Encyclopedia of Philosophy*, written by Simon Friederich.